

Thermal Diffusion Effect on MHD Heat and Mass Transfer Flow Past a Porous Vertical Plate Embedded with Non-Homogeneous P orous Medium

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ABSTRACT This paper studies the thermodiffusion effect on an unsteady flow with heat and mass transfer in an incompressible flow with heat and mass transfer electrically conducting, viscous fluid through a time dependent porous medium past an infinite porous vertical plate with constant suction/injection velocity in the presence of uniform magnetic field applied normal to the flow. The dimensionless governing equations are solved by using perturbation method. The effects of various parameters on primary velocity, secondary velocity, temperature and concentration have been discussed with the help of graphs. While the effects of important parameters on in skin-friction due to primary and secondary velocities, rate of heat and mass transfer have been discussed with the help of tables. When heat absorption parameter increases the primary velocity decreases. Also we noticed that an increase in Soret number results in an increasing primary velocity. A reverse phenomenon is observed on secondary velocity. An increase in Soret number or the exponential index increases the Schmidt number, while Schmidt number increases the rate of mass transfer decreases.

1. INTRODUCTION

In nature, so many flows exist which are caused not only by the temperature differences but also by concentration differences. These mass transfer differences do affect the rate of heat transfer. The phenomenon of heat and mass transfer frequently exists in chemically processes industries such as, food processing and polymer production. The effect of foreign masses, also known as diffusing species concentration, were studied under different conditions by Somers [1], Mathers *et al.* [2] Eckert and Drake [3] have done pioneering work on heat and mass transfer

Many investigations involving convective heat and mass transfer in porous media has been renewed interest in various disciplines such as petroleum reservoirs, coal combustors ground water pollution, filtration processes, chemical catalytic reactors and ceramic process to name just